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AI and Economic Transition | North America

AI and productivity: Productivity growth remains concentrated in high-AI exposed industries

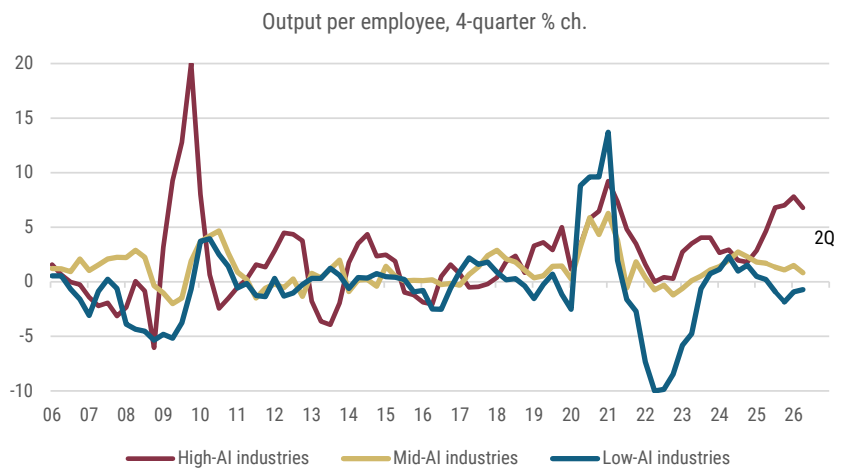
To better assess the effects of AI, our recent publications examined innovation waves, labor displacement risks, and AI productivity. We found stronger productivity in high-AI industries than others. Here, updating, we find continued output & productivity gains, but with some labor replacement.

Key Takeaways

- Productivity, measured as output/employee, is rising faster in industries Morgan Stanley identifies as high-AI exposure than in median- or low-AI industries.
- Productivity accelerated more in high AI exposure industries than in median- or low-exposure industries in 2025 and remains stronger.
- The productivity pickup reflects faster output growth & capital deepening, but now also some labor displacement.

Tech Diffusion
A Morgan Stanley Research Key Theme of 2026

Exhibit 1: In industries with high AI exposure, output per employee has risen faster and accelerated more than in other industries



Source: BEA, BLS, Morgan Stanley Research

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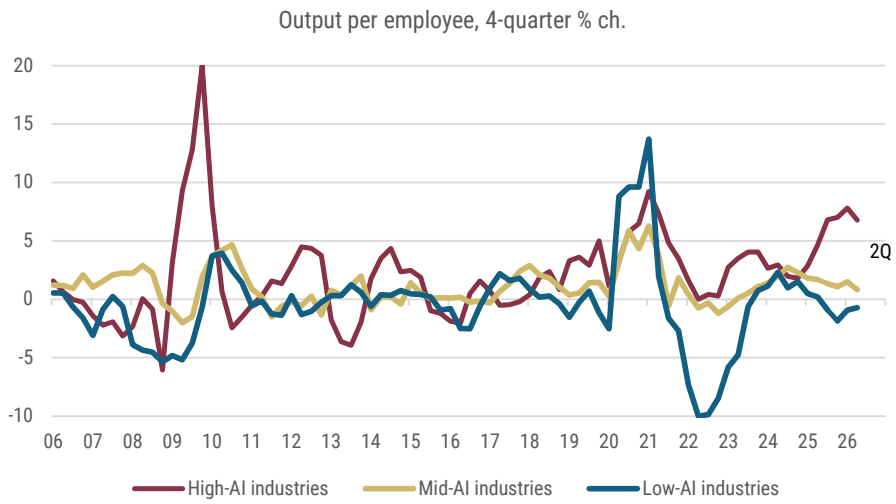
AI-intensive industries drove strong productivity through 2Q

To better understand the effects of AI on growth and the labor market, we examine industry output and employment, classifying industries as having high-, mid-, or low-exposure to AI. We first did this six months ago, finding that high-AI industries had distinctly faster productivity growth than other industries (Please see: Industries with high AI exposure are seeing labor productivity from faster output growth, not labor replacement). Output per employee in these industries had accelerated more than others, and the productivity gains resulted from faster output growth rather than labor replacement. The industry composition of the output gains suggested that gains came from AI-infrastructure investment: capital deepening in AI-related industries rather than business-process changes in the broader economy.

In this report, we update our initial findings. With two more quarters of data in hand, including revised industry output in the GDP annual revision, the contribution of high-AI exposed industries to overall productivity is even greater, and the acceleration in high-AI industry output even larger. But it also now appears as if there may be labor replacement occurring in high-AI industries. This is, so far, a small effect and we characterize it as more micro than macro (see AI labor disruption: Still modest but increasingly visible). Over the past year, the unemployment rate has fallen slightly, the payrolls pace has picked up a bit: the labor market has been strong enough to absorb employment strains from the industries and age groups more exposed to AI. In total, productivity gains continue to come through faster output. Still, labor replacement may be 'micro', but it is a more evident threat than it was six months ago.

Output/employee has risen 2.4%y/y through 2Q. That's the same pace as in 2Q 2025 and below the 3.0%y/y of 1Q 2026. However, it's extremely strong: twice the average pace of the past twenty years. (Our separate decomposition of potential growth into its component parts also continues to point to faster growth in productivity. The 2.1% growth in real GDP over the past four quarters is close to our estimate of potential growth; the strength reflects a supply-side impulse from faster productivity growth rather than overheating demand.)

Exhibit 2: In industries with high AI exposure, output per employee has risen faster and accelerated more than in other industries



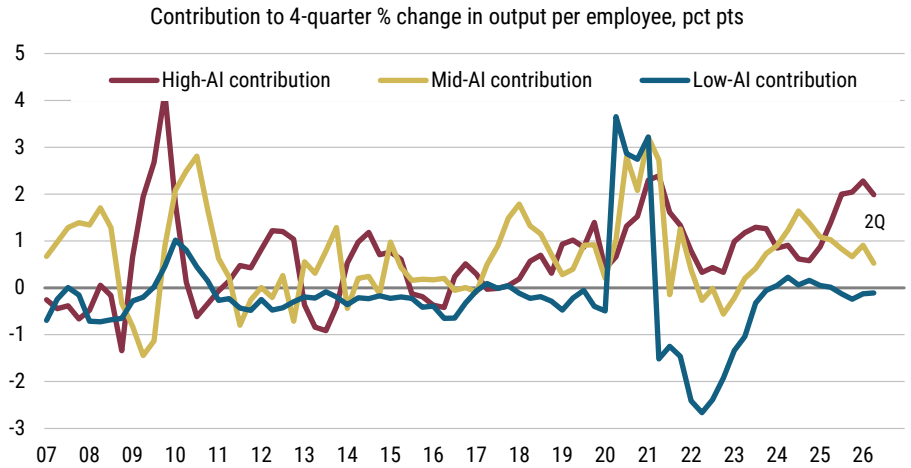
Source: BEA, BLS, Morgan Stanley Research

High AI industries continue to lead the surge in productivity

Industries with high AI exposure account for nearly the entire increase in output per employee. Over the past four quarters, they contributed 2.0pp of the 2.4% rise in economy-side productivity. Mid-AI and low-AI exposure industries together contributed a more modest 0.4pp.

Moreover, the share of the contribution of industries with high AI exposure to overall productivity is increasing. A year earlier, high-AI related industries contributed 1.4pp while mid- and low together contributed 1.0pp. The acceleration in the high-AI industry output/employee from its earlier pace began in late-2024, such that by 2Q 2025 the four-quarter contribution was noticeably above trend, and it has continued since then.

Exhibit 3: Of the 2.4% rise in output/employee over the past four quarters, high-AI industries contributed 2.0pp. The contribution picked up from 1.4pp a year earlier. Mid- and low-AI industries' contributions have been diminishing.



Source: BEA, BLS, Morgan Stanley Research

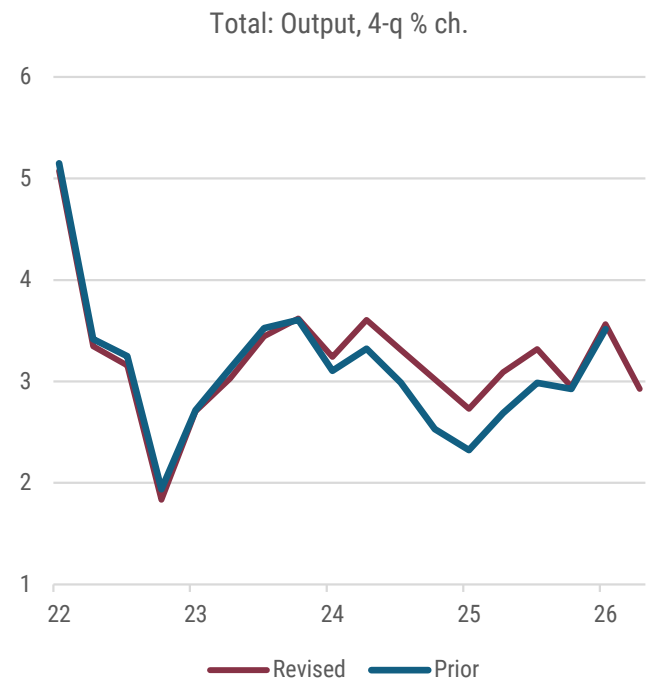
These estimates are based on the GDP data that include this year's annual revisions. Our earlier estimates showed 0.3-0.5pp slower growth in output and output per employee in 2023-24. The newly reported pickup in output per employee in 2023 seems too soon to reflect AI. And indeed, the upward revisions to output were in the mid-AI category rather than the high-AI category. The further strengthening in 2025-26 reflects a new impulse from the high-AI industries. We suspect the earlier gains in productivity reflected efficiency enhancements from the pandemic, which pushed unit labor and non-labor costs higher and incentivized cost control measures.

Exhibit 4: Output per employee: revised up from late-2023 on



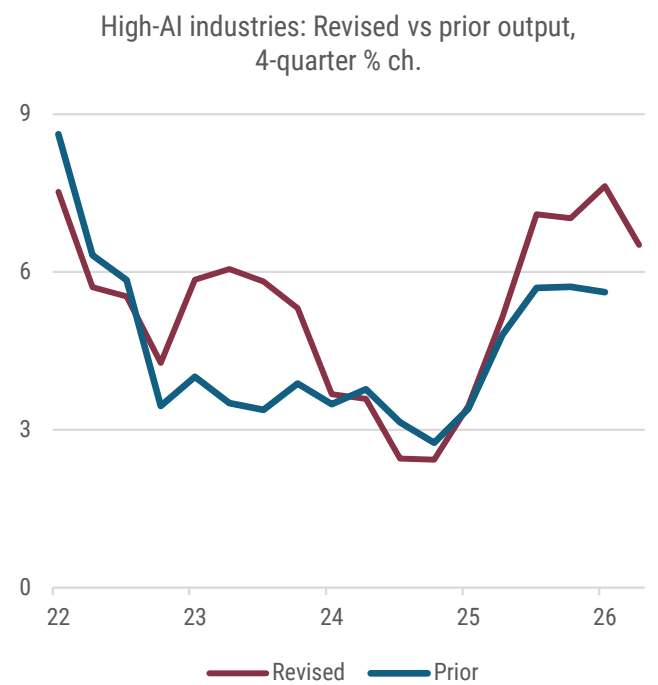
Source: BEA, BLS, Morgan Stanley Research

Exhibit 5: The revision reflected faster output growth



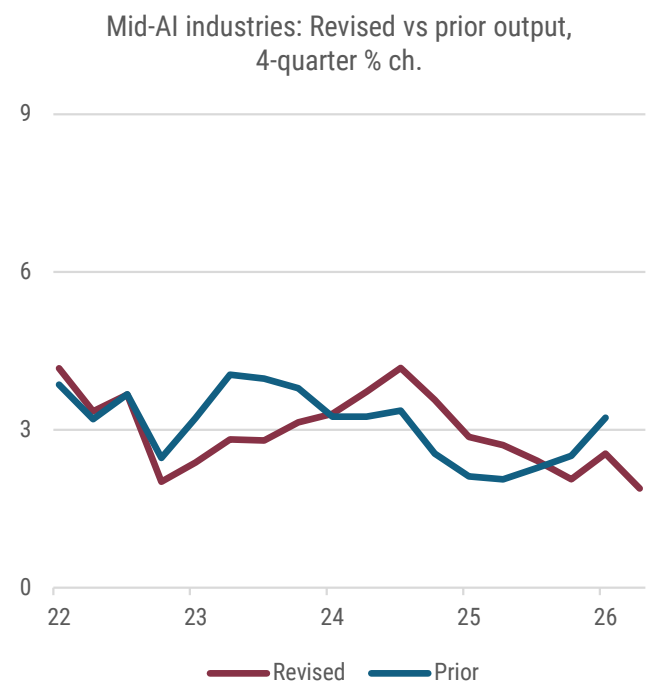
Source: BEA, BLS, Morgan Stanley Research

Exhibit 6: High-AI output: Stronger over the past year after revision



Source: BEA, BLS, Morgan Stanley Research

Exhibit 7: Mid-AI output: Softer over the past year after revision, but stronger before then

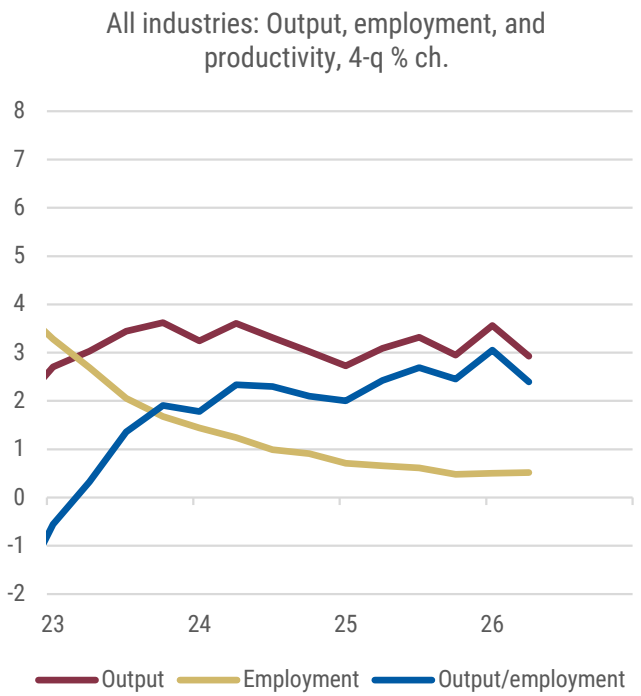


Source: BEA, BLS, Morgan Stanley Research

Labor replacement in high-AI industries?

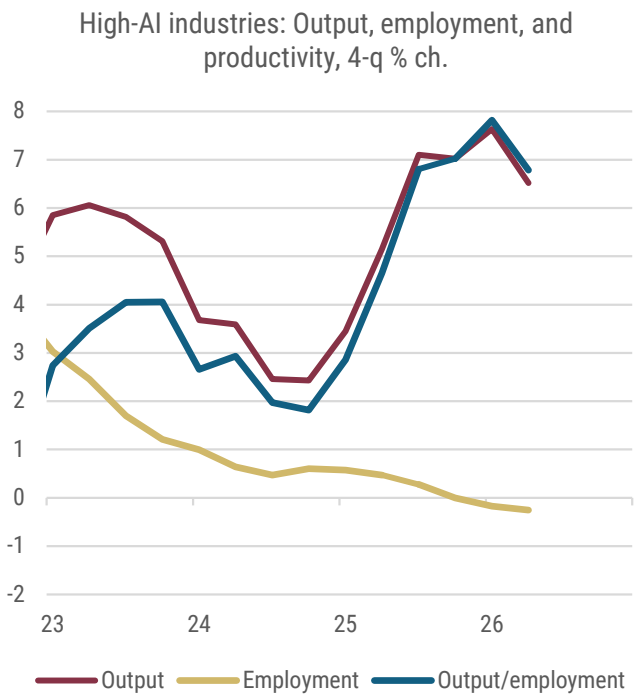
In the latest data, there is some labor replacement that we did not see six months ago. In high-AI industries, employment is now declining, contributing more to productivity growth than it had been. The downturn suggests that AI is creating some labor replacement, and our separate research on AI-related unemployment has shown upward pressure on unemployment for the young in occupations with high exposure to AI (see AI labor disruption: Still modest but increasingly visible).

Exhibit 8: Employment growth has stabilized across all industries after slowing in 2023 and 2024, while output growth has accelerated



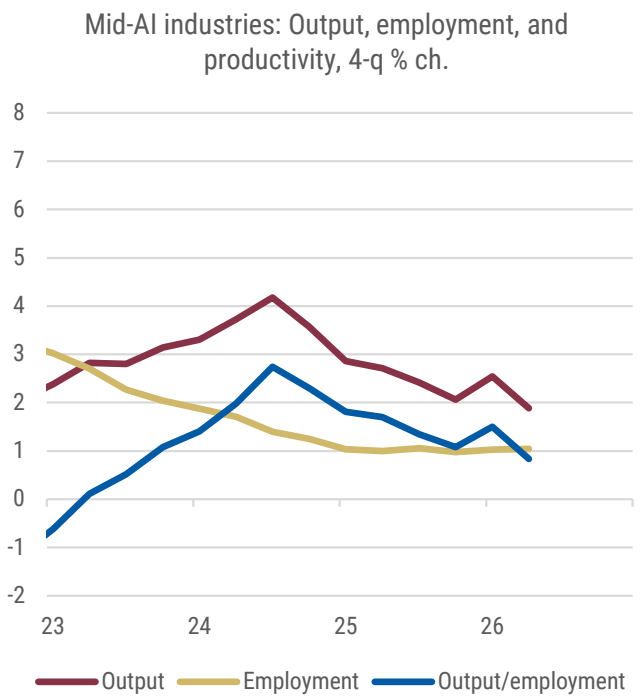
Source: BEA, BLS, Morgan Stanley Research

Exhibit 9: In high-AI industries, the acceleration in output/employee is mostly due to faster output growth, but less employment is contributing, too



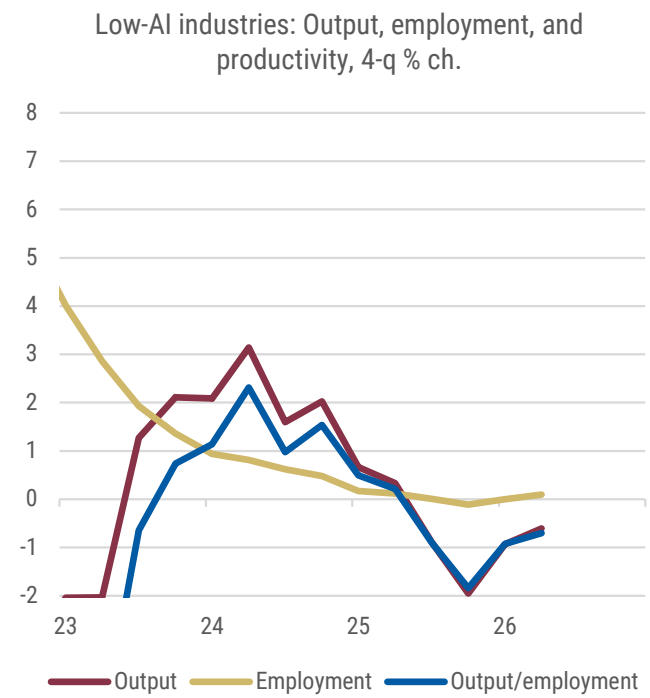
Source: BEA, BLS, Morgan Stanley Research

Exhibit 10: In mid-AI industries, stalled employment growth and slowing output growth



Source: BEA, BLS, Morgan Stanley Research

Exhibit 11: In low-AI industries, slightly less weakness in output growth, stalled employment growth



Source: BEA, BLS, Morgan Stanley Research

So what can we say about labor displacement in high-AI exposed industries? First, about half of the industries we have identified as high AI-exposed are seeing declines in employment. That said, we do not see a clear theme emerging from the downsizing in employment in these industries.

High-AI industries where labor replacement seems likely:

- **Data processing and software:** These industries are losing jobs at an increasing rate amid rapid gains in output.
- **Credit intermediation** is also losing jobs increasingly quickly, with weakness in commercial banking employment and mortgage-related employment. A portion of this could be related to AI adoption, but it may just echo a weak housing market.

High-AI industries where employment weakness seems less likely to be attributable to AI:

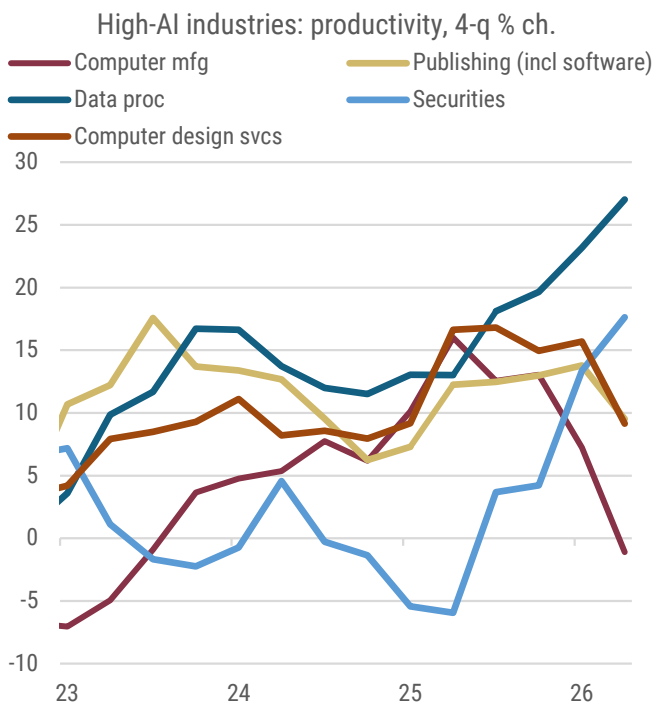
- **Private education** payrolls are falling increasingly quickly but perhaps not because of AI exposure. The sector faces funding threats from increased federal oversight and a shrinking youth population.
- Payrolls in **management** have also stalled after prior gains, but their stall may idiosyncratic after the turmoil of the last year.

High-AI industries where payrolls are rising:

- One industry clearly tied to AI infrastructure that is accelerating its hiring is **computer manufacturing**, but its output and productivity both fell.
- And several other industries also don't look like AI labor replacement: Two are just independently strong: the **securities industry** has rapidly rising output and payrolls rising increasingly quickly. "**Miscellaneous professional and technical services**", which span a wide range, is also strong. And one is weak: **legal services**, where output has stalled yet payrolls are rising.

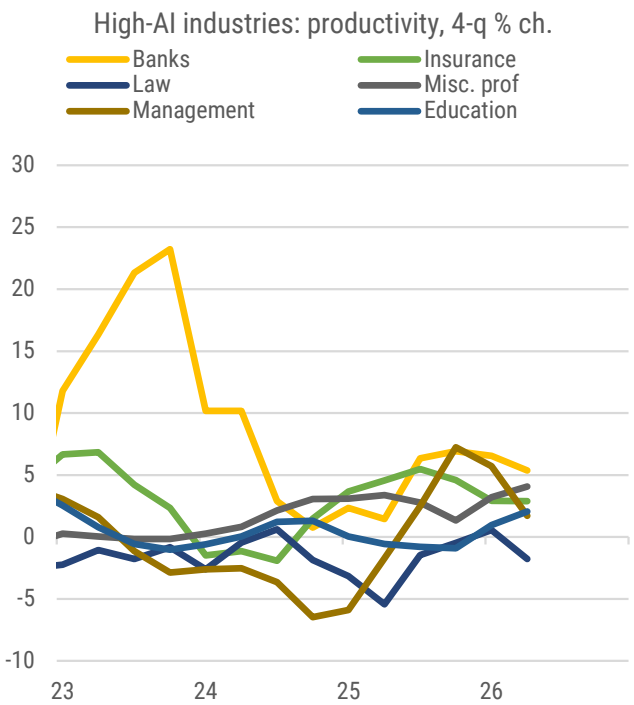
In summary, there's likely some labor replacement in a couple of high-AI industries; but some of the employment decline in other high-AI industries seems equally likely to have causes other than AI; and some high-AI industries show increasing strength in employment. The predominant story to strong productivity growth in high-AI industries is faster output growth. Labor displacement is playing a distant secondary role.

Exhibit 12: Accelerating productivity growth in data processing and securities industries, but slowing in computer manufacturing



Source: BEA, BLS, and Morgan Stanley Research

Exhibit 13: A number of high-AI industries, perhaps less directly tied to the AI infrastructure buildout



Source: BEA, BLS, and Morgan Stanley Research

Methods: Constructing industry-level productivity data

To derive our estimates of labor productivity, we construct industry-level measures of output per employee using the BEA's industry output data and BLS payrolls. Output per employee is an imperfect measure of labor productivity and is not the same as output per hour, which is the preferred measure. While we would prefer to use output per hour, we

believe the official data on nonfarm business productivity and our measure have a strong enough link to enable us to draw inferences about productivity growth at the industry level. (We address the differences between the two measures here: Comparing our productivity measure to nonfarm business productivity .

With calculated industry-level output per employment in hand from 1Q 2006 through 2Q 2026, we then assign each a measures of AI exposure using the criterion in Felten et. al. (2021). [1] . The authors identify occupational exposures to AI and weight those occupations across NAICS 4-digit industry categories. From there, we grouped various industries into three categories: top-25% AI exposure, middle-50% AI exposure, and lowest-25% AI exposure.

In total, we compute output per employment in 64 industries, with 12 industries with high exposure to AI and 26 industries with low exposure to AI. The remaining 26 industries are classified as having medium AI exposure. We then compare and contrast the evolution of output per employee.

Industry output is published quarterly with the third GDP release, and industry payrolls come monthly from the Employment Situation report. This update uses data through 2Q 2026, including the annual GDP revision.

1. https://sms.onlinelibrary.wiley.com/doi/full/10.1002/smj.3286

Exhibit 14: AI exposure and productivity growth, by industry

	AI-exposure	Productivity, 4Q % ch				Accel/Deceleration
	AIIE score	2024 4Q	2025 4Q	2026 1Q	2026 2Q	2Q 2026 *minus* 4Q 2024
Nonfarm business productivity		2.6	2.4	2.8	2.3	-0.3
Employment/output productivity proxy		2.1	2.5	3.0	2.4	0.3
High AI exposure industries (top 25% AIIE)	> 1.25	1.8	7.0	7.8	6.8	5.0
Medium AI exposure industries (middle 50% AIIE)	-0.30 to 1.25	2.3	1.1	1.5	0.8	-1.5
Low AI exposure industries (bottom 25% AIIE)	< -0.30	1.5	-1.8	-0.9	-0.7	-2.2
Securities, commodity contracts, and investments	2.2	-1.4	4.2	13.4	17.6	19.0
Legal services	2.2	-1.9	-0.5	0.6	-1.8	0.1
Credit intermediation, Federal Reserve banks, and related activities	2.1	0.7	6.9	6.5	5.4	4.6
Insurance carriers and related activities	2.1	1.5	4.6	2.9	2.9	1.4
Funds, trusts, and other financial vehicles	2.1	139.2	45.8	37.4	66.7	-72.5
Computer systems design and related services	1.9	8.0	15.0	15.7	9.1	1.2
Publishing industries, except internet (includes software)	1.8	6.2	13.0	13.8	9.5	3.3
Data processing, internet publishing, and other information services	1.7	11.5	19.6	23.2	27.0	15.5
Management of companies and enterprises	1.7	-6.5	7.2	5.7	1.7	8.2
Mfg: Computer and electronic products	1.6	6.2	13.0	7.2	-1.1	-7.3
Miscellaneous professional, scientific, and technical services	1.6	3.1	1.3	3.2	4.1	1.0
Educational services	1.3	1.3	-0.9	1.0	2.1	0.8
Broadcasting and telecommunications	1.2	4.7	1.6	8.0	7.0	2.2
Social assistance	0.8	5.0	-2.2	-2.2	-1.0	-6.1
Ambulatory health care services	0.8	0.5	2.6	2.0	0.9	0.5
Oil and gas extraction	0.7	-5.6	15.9	17.1	0.8	6.4
Hospitals	0.5	1.1	-0.2	1.2	0.3	-0.8
Real estate	0.5	3.3	2.7	2.0	2.4	-0.9
Transit and ground passenger transportation	0.4	16.8	6.4	6.1	5.0	-11.8
Motion picture and sound recording industries	0.4	8.8	-0.3	1.6	0.2	-8.6
Performing arts, spectator sports, museums, and related activities	0.3	6.5	-1.9	0.3	3.8	-2.6
Miscellaneous manufacturing	0.2	-0.3	0.7	3.1	5.3	5.6
Utilities	0.2	0.7	-1.4	0.0	3.7	3.0
Wholesale trade	0.2	2.1	8.1	5.3	3.4	1.3
Mfg: Electrical equipment, appliances, and components	0.2	2.2	5.0	4.0	6.3	4.1
Other retail	0.0	6.0	1.5	0.2	-0.5	-6.6
General merchandise stores	0.0	4.2	-3.8	-14.3	-11.8	-16.0
Other transportation and support activities	0.0	-7.4	-13.8	-6.9	4.1	11.5
Mfg: Chemical products	0.0	12.1	0.1	7.3	0.4	-11.7
Nursing and residential care facilities	0.0	0.4	-1.1	0.2	0.4	0.0
Machinery	-0.1	-6.9	6.9	8.1	11.5	18.4
Pipeline transportation	-0.1	3.4	0.8	2.3	12.6	9.2
Rental and leasing services and lessors of intangible assets	-0.1	10.1	4.8	4.3	3.1	-7.1
Mfg: Other transportation equipment	-0.2	-1.0	15.0	13.8	9.2	10.2
Mfg: Petroleum and coal products	-0.2	0.4	-19.3	-5.6	-18.0	-18.3
Mfg: Printing and related support activities	-0.2	1.6	-0.4	-0.3	-1.2	-2.8
Mfg: Motor vehicles, bodies and trailers, and parts	-0.2	4.9	10.1	15.7	7.8	2.9
Other services, except government	-0.3	0.2	-5.2	-2.2	-1.3	-1.5
Motor vehicle and parts dealers	-0.3	32.9	-2.0	6.8	16.0	-17.0
Air transportation	-0.4	-0.1	-0.4	-3.6	-6.3	-6.2
Food and beverage stores	-0.4	-1.5	-2.6	-1.7	-4.7	-3.2
Administrative and support services	-0.6	3.2	0.4	-0.4	0.3	-2.9
Rail transportation	-0.6	-4.5	-8.9	-5.0	3.8	8.3
Water transportation	-0.6	0.8	3.5	8.1	-7.6	-8.4
Construction	-0.7	0.9	-0.1	0.5	0.0	-0.9
Fabricated metal products	-0.7	-5.3	3.2	8.8	12.2	17.5
Mfg: Apparel and leather and allied products	-0.8	8.6	4.2	7.3	7.0	-1.6
Mfg: Paper products	-0.9	-3.3	-0.2	2.1	2.4	5.7
Accommodation	-0.9	0.5	-1.8	0.9	-2.7	-3.2
Textile mills and textile product mills	-0.9	3.6	1.0	3.3	5.5	2.0
Mfg: Plastics and rubber products	-0.9	-5.4	7.6	9.1	5.1	10.5
Primary metals	-0.9	6.5	-4.9	-6.0	5.5	-1.0
Amusements, gambling, and recreation industries	-1.0	-1.4	0.4	3.2	-0.6	0.7
Nonmetallic mineral products	-1.0	-4.1	-1.1	0.0	-0.9	3.2
Support activities for mining	-1.0	11.6	3.5	-6.0	-6.7	-18.3
Mfg: Furniture and related products	-1.0	1.1	1.9	5.2	4.2	3.0
Waste management and remediation services	-1.0	-0.3	-4.8	-3.5	-2.8	-2.6
Mining, except oil and gas	-1.0	-1.2	-3.1	-1.6	-3.6	-2.4
Mfg: Food and beverage and tobacco products	-1.0	4.0	-7.5	-9.2	-3.2	-7.3
Wood products	-1.1	1.9	2.6	2.9	0.2	-1.7
Food services and drinking places	-1.2	-0.6	-0.8	0.1	0.9	1.5
Truck transportation	-1.3	15.4	-6.7	-1.7	-11.4	-26.8
Forestry, fishing, and related activities	-1.4	3.5	2.7	7.6	6.6	3.1
Warehousing and storage	-1.6	-0.3	-1.8	-0.9	-0.5	-0.1

Source: Morgan Stanley Research. AI-exposure scores are derived from Felten et. al. (2021), https://sms.onlinelibrary.wiley.com/doi/full/10.1002/smj.3286.

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